A3.2 Classification and cladistics (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 LIST the seven ranks of the traditional hierarchy, from kingdom to species. [A3.2.2] [2]

2 DEFINE clade. [A3.2.4] [1]

3 NAME the three domains. [A3.2.9] [1]

4 NAME the molecule whose base sequences led to the three-domain system. [A3.2.9] [1]

5 STATE what a node on a cladogram represents. [A3.2.7] [1]

6 STATE one reason why the molecular clock gives only an estimate. [A3.2.5] [1]

Section B: Understanding

14 MARKS

7 EXPLAIN why the traditional hierarchy of taxa does not always match evolutionary relationships. [A3.2.2] [3]

8 EXPLAIN why organisms are classified, and the advantage of basing classification on evolutionary relationships. [A3.2.1, A3.2.3] [3]

9 **EXPLAIN** how differences in sequences can be used to estimate when two clades diverged. [A3.2.5] [3]

10 **EXPLAIN** why base sequences are stronger evidence for clades than morphology. [A3.2.4] [2]

11 **OUTLINE** how cladistics led to the reclassification of the figwort family. [A3.2.8] [3]

Section C: Working with evidence

12 MARKS

The data in this section are simplified for practice. The pattern matches real findings.

12 [A3.2.5, A3.2.6, A3.2.7]

The table shows the number of differences in the amino acid sequences of a set of proteins between five apes.

	HUMAN	CHIMPANZEE	GORILLA	ORANGUTAN	GIBBON
HUMAN	–	1	2	5	7
CHIMPANZEE		–	2	5	7
GORILLA			–	5	7
ORANGUTAN				–	7

a IDENTIFY the two most closely related species. [1]

b CONSTRUCT a cladogram for the five species. [3]

c CALCULATE when humans and gorillas diverged, if the orangutan lineage split off 15 million years ago. [2]

d SUGGEST two reasons why your answer to c may be inaccurate. [2]

e DEDUCE whether chimpanzees, gorillas and orangutans, without humans, form a clade. [2]

f EXPLAIN why many proteins were compared rather than just one. [2]

Section D: Longer answer

7 MARKS

13 **EXPLAIN** how evidence from rRNA led to the three-domain system, and compare the three domains. [7]

[A3.2.9]